

# Track 2 Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

## Executive Verdict

This update accepts the completed `wave_2c` residual harmonic temporal hybrid campaign into the official `Track 2` offline verification package.

### Decision

- `wave_2c` is verified as an exploratory residual harmonic temporal baseline.
- No `wave_2c` candidate is promoted over the current `wave_2b` periodic sequence leaders.
- The accepted paper-derived forward baseline remains `paper_retuned_best_Fw`.
- The accepted paper-derived backward baseline remains `paper_retuned_best_Bw`.
- The strongest repository-owned static `Track 2` baseline remains `tree`.
- The strongest repository-owned neural branch remains `wave_2b`  
`periodic_gru_sequence_Bw` for backward-only evaluation and `periodic_gru_sequence_global` for bidirectional neural evaluation.

## Source Package

This official report consolidates these refreshed artifacts:

- metric matrix:  
`doc/reports/analysis/track2/Track 2 Directional Model Comparison.md` ;
- matrix summary:  
`output/validation_checks/track2_reference_comparison/2026-05-28-12-22-56__track2_full_directional_family_matrix_wave2c_residual_harmonic_temporal_hybrid_track2_refresh_2026_05_28/validation_summary.yaml` ;
- per-condition metrics:  
`output/validation_checks/track2_reference_comparison/2026-05-28-12-22-56__track2_full_directional_family_matrix_wave2c_residual_harmonic_temporal_hybrid_track2_refresh_2026_05_28/per_condition_metrics.csv` ;
- best-model collage report:  
`doc/reports/analysis/track2/best_model_collage_report/[2026-05-28]/track2_best_model_collage_report.md` ;
- multi-model curve comparison report:  
`doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-05-28]/track2_multi_model_curve_comparison_report.md` .

## Candidate Refresh

The refresh added 18 registry-backed wave 2C candidates:

| Family                                      | Surfaces         |
|---------------------------------------------|------------------|
| residual_harmonic_gru_sequence_sparse_rcim  | global , Fw , Bw |
| residual_harmonic_gru_sequence_dense240     | global , Fw , Bw |
| residual_harmonic_gru_sequence_dense360     | global , Fw , Bw |
| residual_harmonic_lstm_sequence_sparse_rcim | global , Fw , Bw |
| residual_harmonic_lstm_sequence_dense240    | global , Fw , Bw |
| residual_harmonic_lstm_sequence_dense360    | global , Fw , Bw |

The matrix now contains 111 candidates. The incremental operator run used the completed wave 2B refresh as its 93 -candidate baseline and evaluated only the 18 new wave 2C candidates.

## Current Leaders

| Scope                      | Current strongest candidate  | MAE [deg] | RMSE [deg] | Mean [%] |
|----------------------------|------------------------------|-----------|------------|----------|
| forward overall            | rcim_retuned_GBM19_Fw        | 0.001089  | 0.001299   | 2.372    |
| backward overall           | periodic_gru_sequence_Bw     | 0.002392  | 0.002639   | 5.466    |
| paper-derived forward      | paper_retuned_best_Fw        | 0.001839  | 0.002041   | 4.109    |
| paper-derived backward     | paper_retuned_best_Bw        | 0.003675  | 0.004284   | 7.572    |
| repository static backward | tree_Bw                      | 0.003258  | 0.003651   | 7.051    |
| repository global static   | tree_global                  | 0.003144  | 0.003533   | 6.854    |
| repository global neural   | periodic_gru_sequence_global | 0.002704  | 0.002949   | 6.139    |

## Wave 2C Result

The strongest wave 2C candidates are

| Scope    | Strongest Wave 2C candidate                        | MAE [deg] | RMSE [deg] | Mean [%] |
|----------|----------------------------------------------------|-----------|------------|----------|
| forward  | residual_harmonic_gru_sequence_sparse_rcim_Fw      | 0.003194  | 0.003499   | 7.083    |
| backward | residual_harmonic_lstm_sequence_sparse_rcim_Bw     | 0.003440  | 0.003793   | 7.510    |
| global   | residual_harmonic_lstm_sequence_sparse_rcim_global | 0.003368  | 0.003719   | 7.409    |

The sparse RCIM harmonic set is clearly the best wave 2C setting. The dense 240 and dense 360 residual harmonic variants are not competitive in the official matrix and appear to over-expand the harmonic base for this residual sequence branch.

## Visual Evidence

The refreshed best-model collage report includes dedicated wave 2C sections for forward, backward, and global candidates. The refreshed multi-model overlay report includes dedicated wave 2C overlays and direct reference/tree versus wave 2C overlays.

Visual bundle paths

- doc/reports/analysis/track2/best\_model\_collage\_report/[2026-05-28]/track2\_best\_model\_collage\_report.pdf ;
- doc/reports/analysis/track2/multi\_model\_curve\_comparison\_report/[2026-05-28]/track2\_multi\_model\_curve\_comparison\_report.pdf .

## Campaign Update Ledger

| Date       | Campaign or Update                                | Candidate Scope                                                              | Matrix Status                                             | Visual Status                                               | Decision                                                                               |
|------------|---------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 2026-05-28 | wave 2C residual harmonic temporal hybrid refresh | 18 global , Fw , and Bw residual harmonic GRU/LSTM candidates                | included in the 111 - candidate matrix                    | collage and overlay reports refreshed with wave 2C sections | verified exploratory baseline; not promoted over wave 2B or accepted Track 2 baselines |
| 2026-05-26 | wave 2B harmonic temporal hybrid refresh          | periodic temporal convolution, GRU, and LSTM global , Fw , and Bw candidates | included                                                  | collage and overlay reports refreshed                       | strongest repository-owned neural branch                                               |
| 2026-05-24 | wave 2 temporal entry refresh                     | temporal convolution, GRU, and LSTM global , Fw , and Bw candidates          | included                                                  | visual reports refreshed                                    | verified exploratory baseline                                                          |
| 2026-05-21 | periodic_mlp explicit harmonic registry refresh   | global , Bw visual verification                                              | source matrix unchanged; visual registry refresh included | collage and overlay PDFs refreshed                          | included as visual evidence, not promoted over tree                                    |

## Closeout Decision

wave 2C does not change the accepted Track 2 baseline. Its practical value is diagnostic: it confirms that adding a residual recurrent branch over the structured harmonic base is viable only with the sparse RCIM harmonic set, while the dense harmonic banks are dimensionally expensive and less accurate.

The next modeling decision should keep wave 2B periodic sequence models as the strongest neural branch and treat wave 2C as a verified exploratory baseline rather than the next promotion target.